

Jeet Pancholi

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EDUCATION

University of Illinois Urbana-Champaign | Champaign, IL

Expected December 2027

Bachelor of Science in Mathematics and Computer Science

GPA: 3.74/4.0

Relevant Coursework: Linear Algebra, Calculus III, Computer Systems, Data Mining, Data Structures, Statistics & Probability

EXPERIENCE

Kripa Montessori School

Schaumburg, IL

Full Stack Developer Intern

May 2025 - August 2025

- Developed a React and Tailwind CSS web prototype from scratch as the technical lead for a proof of concept that modernized the school's digital experience and improved usability for staff and families.
- Managed the full development workflow using Git and Visual Studio Code, maintaining clean version control, organizing project structure, and clear documentation that reduced developer handoff friction by an estimated 20 to 25%.
- Organized legacy school data by scanning physical records and creating a metadata tagging system to make their internal archives searchable for the first time, cutting manual record lookup time by an estimated 50 to 60%.

Teaching Aide I, University of Illinois Chicago

Chicago, IL

MCS 160 Teaching Aide I

February 2025 - May 2025

- Streamlined grading for 30+ students by implementing a standardized feedback rubric for Python assignments.
- Reviewed student code for correctness, efficiency, and style, and provided feedback to reinforce course concepts.

Driving Forward Program

Chicago, IL

Artificial Intelligence Consulting Intern

January 2025 - March 2025

- Worked in a consulting team with undergraduate students to assess operational inefficiencies in primary care workflows that reduced time available for direct patient interaction and increased administrative burden on physicians.
- Designed and deployed an unbiased survey of randomly selected primary care physicians to collect quantitative and qualitative data on workflow bottlenecks, documentation burden, and opportunities for automation.
- Proposed an AI-driven automated scribe solution projected to reduce administrative overhead by 15 to 20% and increase time available for patient interaction by redirecting physician effort away from repetitive documentation tasks.

EXTRACURRICULAR ACTIVITIES

Illinois Data Science Club

Urbana, IL

Apple Options Volatility Prediction

September 2025 - December 2025

- Analyzed 2,000+ Apple option contracts using implied volatility, days to expiration (DTE), strike price, and trading volume to identify key drivers of option pricing and improve feature selection for predictive modeling.
- Applied TextBlob sentiment analysis to X/Twitter data and integrated sentiment signals with options market data to evaluate how shifts in public sentiment aligned with short-term changes in implied volatility.
- Built and evaluated machine learning models to predict implied volatility, with a Random Forest model achieving $R^2 = 0.97$, demonstrating strong predictive accuracy and effectively capturing non-linear relationships across market variables.

ACADEMIC PROJECTS

Asynchronous Image Flood Fill Engine

Fall 2025

- Developed a high-performance image processing tool in C++ using Breadth-First Search (BFS) and Depth-First Search (DFS) to execute multi-directional flood-fill operations on high-resolution PNG datasets.
- Architected custom C++ Iterators to abstract complex pixel traversal logic, providing a modular framework that supports extensible image transformations and dynamic animation sequences across thousands of data points.
- Optimized system-level memory allocation by implementing localized stacks and queues, reducing peak memory overhead by an estimated 25 to 35% and enabling real-time rendering of 100+ frame high-fidelity GIF animations.

CookingPal | Pantry Organization & Recipe Recommendation Mobile Application

Summer 2025

- Developed a cross-platform mobile application using React Native and Expo Go to support pantry inventory management.
- Implemented manual and image-based grocery item entry using Google Cloud Vision API for handwritten text recognition.
- Integrated Firebase for secure cloud-based data storage, enabling dynamic updates and user-specific data management.
- Leveraged RapidAPI to fetch food images and generate personalized recipe recommendations based on pantry items.

SKILLS AND TECHNICAL TOOLS

Languages & Frameworks: Python, JavaScript, C, C++, HTML, CSS, React, React Native, Flask, JavaFX, Tailwind CSS

Libraries & Technologies: NumPy, Pandas, Matplotlib, REST APIs, Firebase, SQLite

Tools & Platforms: Git, GitHub, VS Code, Docker, Linux, Unix, Jupyter Notebook, Expo Go